

Inventory Management Business Analysis Project

End-to-End BA Case Study & Power BI Implementation

Objective: Analyze inventory risks, improve replenishment decisions, identify slow-moving products, and provide management with actionable insights through Excel and Power BI.

1. Business Problem

The business faced assumption-based reordering, manually prepared inventory reports, delayed reorder decisions, limited timely inventory information, and poor visibility into slow-moving products. The goal was to create a data-driven inventory monitoring and decision-support solution.

2. Stakeholders

Stakeholder	Primary concern
Senior Management	Profitability, inventory performance, decisions
Finance	Inventory value, working capital, efficiency
Production	Production and replenishment needs
Sales	Product availability and demand
Inventory Manager	Stock levels, reorder points, slow-moving inventory
Wholesale & Retail	Product availability and demand

3. Key Business Questions

- Which products have the highest demand?
- Which products become unavailable when customers need them?
- How should reorder decisions be made?
- Which products remain in inventory for long periods?
- What are the problems with the current inventory process?

4. Requirements & Solution Direction

Area	Requirement / improvement
Demand Forecasting	Use historical demand data instead of assumptions.
Reorder Point	Review reorder points using demand and replenishment lead time.
Inventory Monitoring	Regularly monitor inventory and provide timely updates.
Slow-Moving Products	Identify products with no sales over the agreed 90-day threshold.
Alerts	Notify responsible teams when inventory reaches the defined reorder condition.

5. Requirement Prioritization

Must Have: Timely inventory/reorder information and core monitoring capability.

Should Have: Reorder-point improvement and slow-moving identification.

Could Have: Additional forecasting and analytical enhancements.

6. User Acceptance Testing

Test	Expected result
UAT-001	When stock is at/below reorder point, generate a reorder alert.
UAT-002	Production receives replenishment notification within the agreed 1-minute target.
UAT-003	Product with no sales for 90 days is classified as slow-moving.
UAT-004	Dashboard KPIs match approved source-data calculations.
UAT-005	Reorder-point changes retain old value, new value, user, date/time and reason where applicable.

7. Excel KPI Results

KPI	Result
Total Inventory Units	1,070
Products Requiring Reorder	0
Stockout Rate	0%
Slow-Moving Inventory	10%
Inventory Value	■5,78,000
Average Inventory Value	■6,14,000
Total COGS	■26,12,800
Inventory Turnover	4.26×

8. Demand & Inventory Findings

Classic Cotton Shirt: 638 units sold in 90 days; highest demand and about 17 days of stock coverage.

Kids Jeans: 0 sales in 90 days with 150 units in stock; slow-moving and potential excess inventory.

Running T-Shirt: 6 sales in 90 days; about 2 units/month and approximately 525 days of stock coverage.

Women Ethnic Dress: 144 units sold in 90 days; not slow-moving but should be monitored for stock risk.

9. Inventory Value by Category

Category	Inventory Value	Share
Apparel	■2,46,750	42.7%
Ethnic Wear	■1,47,000	25.4%
Kids Wear	■1,41,000	24.4%
Sports Wear	■43,250	7.5%
Total	■5,78,000	100%

10. Stockout Risk Analysis

Premium Denim Jeans: 15 days of stock vs 20-day lead time → High Risk.

Sports Track Pants: 22 days of stock vs 25-day lead time → High Risk.

11. Power BI Dashboard

Dashboard sections: KPI Cards; Inventory Value by Category; Demand by Product; Reorder Risk Matrix; Slow-Moving Products; and Key Business Insights.

12. Key Business Recommendations

- Review reorder points using historical demand, replenishment lead time and safety-stock policy.
- Closely monitor Premium Denim Jeans and Sports Track Pants.
- Investigate Kids Jeans because ■75,000 of inventory value is tied up in a product with zero sales over 90 days.

- Review Running T-Shirt because demand is very low and inventory coverage is extremely high.
- Replace manual/assumption-based monitoring with timely, data-driven inventory alerts.
- Compare inventory turnover against company targets or historical performance before judging 4.26x.

13. BA Skills Demonstrated

Business problem analysis • Stakeholder identification • Requirements elicitation • Requirement prioritization • User stories • Acceptance criteria • RTM • UAT • Excel data analysis • KPI design • Demand analysis • Inventory risk analysis • Power BI • DAX • Data-driven business recommendations

14. Project Outcome

The project transformed an inventory-management problem into a structured, data-driven decision-support solution. It identified both potential stockout risk in high-demand products and excess/slow-moving inventory in low-demand products, then translated the findings into a management-focused Power BI dashboard.